

PH Meter Buying Guide

From German brewing and more

When looking for pH meters you'll find that there are a lot of different models with a variety of features and prices ranging from \$50 to \$500 and once you justified the expense you'll ask yourself what meter should I get as a brewer.

pH meters are glorified volt meters that measure the electrical potential produced by a special pH probe. What determines their quality and precision is the quality of the probe and the quality of the amplifier and analog-digital converter. The probe wears over time and you should expect that you'll have to replace it every 2-3 years if you take good care of it. The probe's response to pH is linear. This means there is a linear function between the voltage it produces and the pH of the solution it is in. This linear function has an offset and a slope which are the parameters that are set when you calibrate it. When the probe ages the slope will deteriorate and get flatter. This means you'll get less voltage difference for the same pH difference.

pH meters seem to come in 3 forms: pen-style, portable hand held and bench top. Bench top meters are generally the most precise, least mobile and most expensive.

When it comes to choosing a pH meter, these are the specs and features you should look for:

- **precision:** for brewing purposes you want to have at least ± 0.01 .. 0.02 pH units. This is plenty for testing mash pH and good enough for detecting pH rises in beer that can be a sign of autolysis. Some meters provide only ± 0.1 which I consider too little and you may quickly find that you would like more than that. But if the meter is cheap enough you may go with that precision if you are only interested in testing the mash pH.
- **calibration:** Two concepts exist here: manual and automatic. Manual calibration requires you to adjust 2 knobs (one for the offset (pH 7.00) and one for the slope (pH 4.00 or 10.00)) after you placed the probe into the appropriate calibration buffer. Automatic calibration does this for you. It tells you what buffer to place the probe in and/or recognizes the buffers on its own. My first meter had an automatic calibration and it failed to recognize the buffers fairly early in its life. From that point on I had to do the calibration externally by using the values read for the 4 and 7 pH buffers and the value read for the sample. I'm now a proponent of manual calibration as it is more hands-on and the meter won't be able to refuse to calibrate
- **temperature correction:** pH meters come with no, manual or automatic temperature correction (ATC). Besides the pH of the sample the pH probe is also affected by the temperature of the sample and in order to determine pH the temperature of the sample needs to be known. But the need for temperature compensation is generally overstated since the actual pH of the sample may also change with temperature. The latter requires you to state the temp that the pH was measured at or always measure pH at the same temperature. The same is true for the calibration buffers which have their nominal pH only at 25C and this is the temperature at which you should calibrate the meter. Otherwise you need to look up the pH that the buffer solution has at its current temperature. Many buffer solutions have table that will show you their pH changes with temperature. The same is true for wort. I always measure all my pH values at 25C which means that beer needs to be heated to this temp and wort and mash samples need to be cooled. As a result I do not have a real need for a temperature compensation on the pH meter. pH meters with manual temp compensation will have a dial or other means of entering the sample temperature and ATC meters have a temperature probe in addition to the pH probe. That probe may be in the same housing as the pH probe or a separate probe altogether. The latter allows you to test the sample temp before you submerge the pH probe in a possibly hot sample.

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NOTE: many brewers think that ATC means that you can test the mash pH at any temp within the pH meter's temperature range. While this is true you still need to know the temperature dependent pH shift in order to correct the pH temp to the standard temperature at which the optimal pH targets were published. Briggs and DeClerck cite a pH shift of -0.35 between a room temps and a mash temp sample while my own experiments showed only -0.18. There doesn't seem to be much data about this shift out there and the majority of the pH values that are given for brewing are room temp pH measurements. Just because a meter has ATC it is not more accurate, especially if the sample's temperature to pH function is unknown.

- **pH probe connection:** The standard for connecting pH probes are BNC connections. If you have a meter with a BNC connection you may be able to shop around for replacement probes as you are not tied to a specific connector design. It is possible that incompatibilities still exists that I'm not aware of, so treat this statement with caution. BNC connectors are generally found on bench-top or portable meters but not on the pen-style ones.
- **mV read-out:** Since the slope of a pH meter can be used to asses its age it is nice if the pH meter also has a mV read-out. But I have not seen this on the entry level pH meters that are priced attractive to brewers
- **price:** I think that you can get a decent pH meter suitable for brewing applications between \$50 and \$100. You can pay more but the added benefit may not be worth the money and you may be better off spending this money on another neat piece of testing equipment: A dissolved oxygen meter.

After considering all these facts I personally settled on the SM101 from Milwaukee (<http://www.milwaukee testers.com/SM101.html>)

You can find it for ~\$80 on the web. It has an accuracy of 0.01, manual calibration and BNC probe connector. The temp correction is manual for the few cases where I want to measure non 25C samples. I have been using it for about 4 weeks now and am very pleased with it. After the initial calibration I have noticed an only very slight drift which means that I can trust the pH reading even without constant calibration. I check the calibration whenever I use it extensively but don't calibrate it for occasional uses since I rarely have to change the calibration anyway.

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The Milwaukee SM101 is a reliable pH meter well suited for brewing that cost less than \$100

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